TECHNICAL TRAINING CATALOGUE 2026

SHACMAN

OUR COMMITMENT:

TO ASSIST YOU IN DEVELOPING THE COMPETENCE AND PERFORMANCE OF YOUR EMPLOYEES AND YOUR SERVICE.

Version of October 27, 2025

EDITO

Dear Subsidiary,

The Group Academy is pleased to present its catalogue of services for 2026 and to inform you of our commitments for all subsidiaries:

- ✓ **To support you in developing the skills and performance of your employees and your service,**
- ✓ **To support you in resolving technical problems on customer vehicles,**
- ✓ **To standardize and normalize the skill levels of the group's technicians.**

The diversity of brands deployed by CFAO is a real asset for the development of our skills development offers to subsidiaries throughout the company's territories. This diversity challenges us to ensure a high level of service quality within the after-sales service, the parts service, the sales department and the financial and legal service.

To improve the quality of your employees' services, we recommend deploying the following technical skills acquisition cycle:

- ✓ **Measure** the current skills of technicians and teams,
- ✓ **Exploit** the results to guide the skills development strategy,
- ✓ **Define** individual and collective skills development plans,
- ✓ **Acquire** new skills through face-to-face or remote trainings,
- ✓ **Certify** the new skills acquired,
- ✓ **Apply** new skills in a professional environment,
- ✓ **Perform** in a professional environment.

We also offer to **assist** you in **resolving technical problems** that you encounter in your workshops, with **technical support integrated** into the Academy.

The development of new skills is carried out by certified experts each in their areas of expertise, using innovative training methods.

Thank you for your commitment to developing the skills of your employees. We look forward to hearing from you.

Laurent GONIN

TEAM EQUIPMENT

The Douala Academy for EQUIPMNET is dedicated to the **RENAULT TRUCKS, JCB, LOVOL, TOYOTA FORKLIFT, BT, FUSO, HINO, SINOTRUK, SHACMAN and KING LONG** brands and is equipped with:

- A reception and a restaurant,
- Two training rooms (board, video projector, others),
- A 200 m² workshop,
- Didactic equipment for engines, transmissions and others,
- A RENAULT TRUCKS K Range truck

Our training courses are delivered at the Douala Academy and within all our subsidiaries. We also provide training at our customers' sites. Training sessions are made up of a maximum of twelve trainees.

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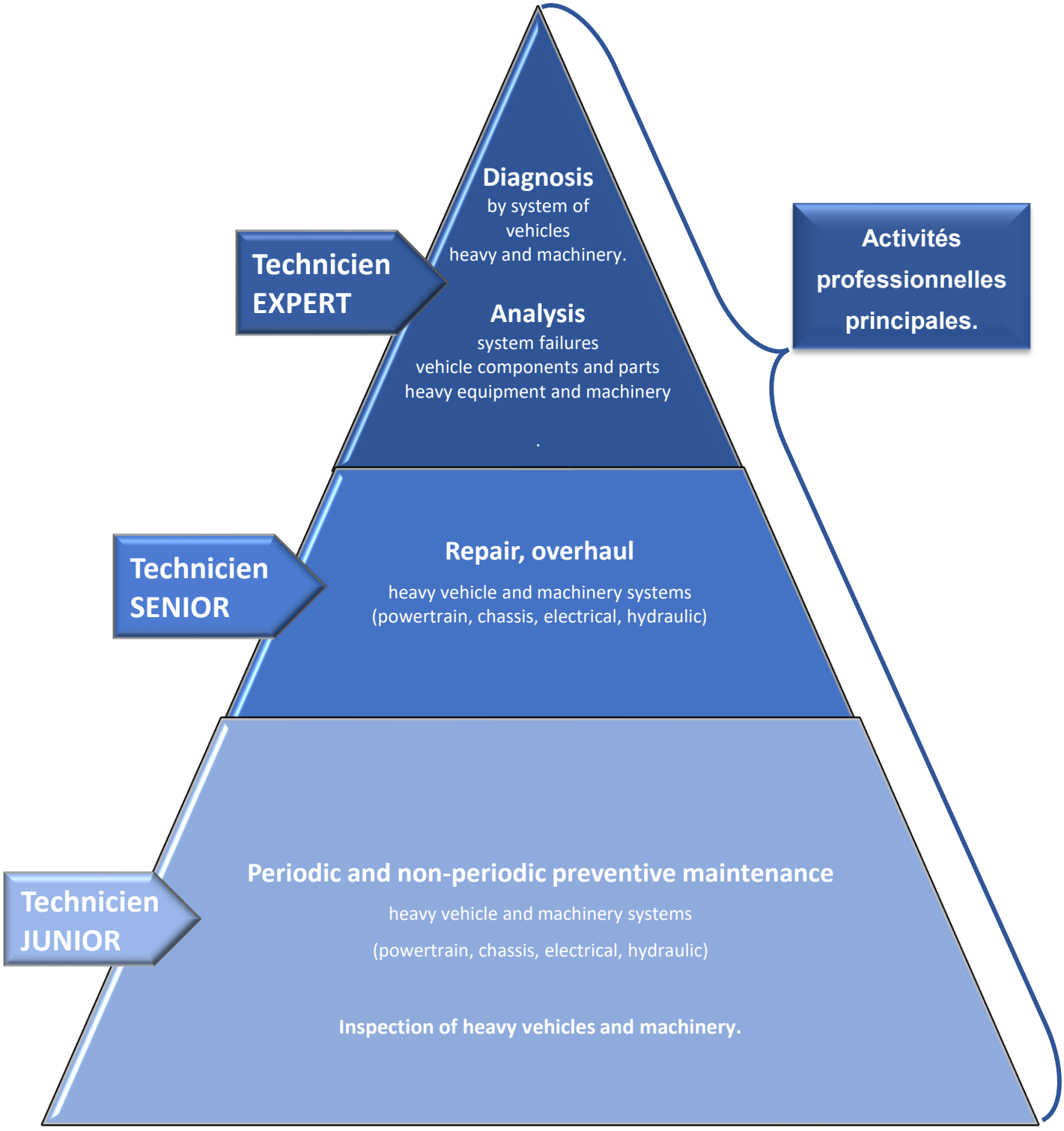


Roland NENDA
Operator Trainer
All- brands expert (Trucks and Heavy Equipment)

DEVELOPING TECHNICIANS' PROFESSIONAL EXPERTISE



SKILL LEVELS OF TECHNICIANS



TRAINING BY TECHNICIAN LEVEL

	Theoretical Training (Knowledge -> E-Learning/Distance Learning)		training (Know-how -> Face-to-face)	
	Themes	Duration (days)	Themes	Duration (days)
Technicien EXPERT	❖ The 7 steps of the diagnostic applied to the different systems of the truck (powertrain, chassis, electrical system), ❖ Diag Tool Features	20	❖ Diagnostics (HOHAN) 5 ❖ HW19710 Gearbox Diagnostics 5 ❖ Electrical System Diagnostics 5 ❖ Conventional braking system diagnostics (with ABS) 5 ❖ Diagnostics for Bridges and Drive Gearboxes, mcy11 series 5	
Technicien SENIOR	❖ Technological knowledge of the different truck systems (powertrain, chassis, electrical). ❖ Removal and installation procedure in accordance with the manufacturer, ❖ Disassembly and reassembly procedure in accordance with the manufacturer ❖ Manufacturer-compliant component inspection procedure	20	❖ WD615 Engine Repair (HOHAN) 5 ❖ HW19710 Gearbox Repair 5 ❖ Electrical System Repair 5 ❖ Conventional braking system repair (with ABS) 5 ❖ mcy11 series drive 5	
Technicien JUNIOR	❖ Brand history (truck ranges, characteristics, etc.), ❖ Description of the truck's structure and the roles of the different truck systems (powertrain, chassis, electrical). ❖ Role of liquids and lubricants. ❖ Principle of use of metrology tools, ❖ Periodic and non-periodic preventive maintenance of trucks, ❖ The fundamental principles of the engine, transmission, electricity, ❖ Others.	25	❖ Preventive maintenance of HOHAN Truck systems	5

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10	S102	Preventive Maintenance - SHACMAN X3000 Range	To be defined	To be defined
11	S103	Preventive maintenance - SHACMAN M3000 range (6x4)	To be defined	To be defined
12 to 17	SENIOR TECHNICIAN (REPAIR)			
13	S201	Cummins Engine Repair (e.g., ISM/ISL)	To be defined	To be defined
14	S202	Gearbox Repair	To be defined	To be defined
15	S203	SHACMAN Electrical System Repair	To be defined	To be defined
16	S204	Conventional braking system repair (with ABS/ASR)	To be defined	To be defined
17	S205	Repair of Bridges and Gearboxes by Hand	To be defined	To be defined
18 to 23	EXPERT TECHNICIAN (DIAGNOSTICS)			
19	S301	Cummins Engine Diagnostics (e.g., ISM/ISL) and Electronic Architecture	To be defined	To be defined
20	S302	Fuller/Fast Gearbox Diagnostics and Controls	To be defined	To be defined
21	S303	SHACMAN Advanced Electrical and Electronic System Diagnostics	To be defined	To be defined
22	S304	Conventional braking system and ABS/ASR diagnostics	To be defined	To be defined
23	S305	Bridge and Gearbox Diagnostics HanDe	To be defined	To be defined

JUNIOR TECHNICIAN

PREVENTIVE MAINTENANCE



S101	Preventive Maintenance of Shacman Truck Systems

Level, Mode and Duration	JUNIOR	Face-to-face	5 Days
Presenter	**SHACMAN Expert Technical Trainer**		
Goals	• **Maintain** systems (powertrain, chassis, electrical) in operation according to the manufacturer's recommendations, • **Inspect** systems (powertrain, chassis, electrical system) of the heavy goods vehicle according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Truck ranges and their characteristics, • Description of the vehicle structure and the role of the different truck systems (Powertrain, Chassis, Electrical) and their components, • Roles of liquids and lubricants, • Principle of use of metrology tools, • The different types of maintenance applied to SHACMAN trucks and maintenance periods in accordance with the manufacturer's specifications, • Preventive maintenance, periodic and non-periodic, of systems and components in accordance with the manufacturer's specifications. ► **Part Practical :** • Replace oils and fluids (Engine, Gearbox, axles, others). • Top up the oils and fluids (engine, gearbox, axles, others) • Remove and reinstall the components (belt and accessories, brake pads or linings, etc.) according to the manufacturer's recommendations. • Check the conformity of the components (dimensions, flatness, voltage, current…) to the manufacturer's expectations before reassembling them. • Visually inspect the truck's systems. ► **Part Assessment :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	Having completed training in basic mechanics and electricity		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, Satisfaction sheet, Certificate of participation.		
Date and Place	To be defined		To be defined

S102	PREVENTIVE MAINTENANCE - RANGE X 3000

Level, Mode and Duration	JUNIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Maintain** systems (powertrain, chassis, electrical) in operation according to the manufacturer's recommendations, • **Inspect** systems (powertrain, chassis, electrical system) of the heavy goods vehicle according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical Part:** • X3000 truck ranges and their characteristics, • Description of the vehicle structure and the role of the different truck systems (Powertrain, Chassis, Electrical) and their components, • Roles of liquids and lubricants, • Principle of use of metrology tools, • The different types of maintenance applied to SHACMAN trucks and maintenance periods in accordance with the manufacturer's specifications, • Preventive maintenance, periodic and non-periodic, of systems and components in accordance with the manufacturer's specifications. ► **Practical Part:** • Replace oils and fluids (Engine, Gearbox, axles, others). • Top up the oils and fluids (engine, gearbox, axles, others) • Remove and reinstall the components (belt and accessories, brake pads or linings, etc.) according to the manufacturer's recommendations. • Check the conformity of the components (dimensions, flatness, voltage, current…) to the manufacturer's expectations before reassembling them. • Visually inspect the truck's systems. ► **Part Assessment :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	Having completed training in basic mechanics and electricity		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, Satisfaction sheet, Certificate of participation.		
Date and Place	To be defined		To be defined

S103	PREVENTIVE MAINTENANCE – M3000 (6X4/8X4)

Level, Mode and Duration	JUNIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Maintain** systems (powertrain, chassis, electrical) in operation according to the manufacturer's recommendations, • **Inspect** systems (powertrain, chassis, electrical system) of the heavy goods vehicle according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Theoretical Part:** • M3000 truck ranges and their characteristics, • Description of the vehicle structure and the role of the different truck systems (Powertrain, Chassis, Electrical) and their components, • Roles of liquids and lubricants, • Principle of use of metrology tools, • The different types of maintenance applied to SHACMAN trucks and maintenance periods in accordance with the manufacturer's specifications, • Preventive maintenance, periodic and non-periodic, of systems and components in accordance with the manufacturer's specifications. ► **Practical Part:** • Replace oils and fluids (Engine, Gearbox, axles, others). • Top up the oils and fluids (engine, gearbox, axles, others) • Remove and reinstall the components (belt and accessories, brake pads or linings, etc.) according to the manufacturer's recommendations. • Check the conformity of the components (dimensions, flatness, voltage, current…) to the manufacturer's expectations before reassembling them. • Visually inspect the truck's systems. ► **Part Assessment :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Operational technicians in preventive maintenance		
Prerequisites	Having completed training in basic mechanics and electricity		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, Satisfaction sheet, Certificate of participation.		
Date and Place	To be defined		To be defined

SENIOR TECHNICIAN

REPAIR



Moteur Cummins



BoîteS de Vitesses FULLER/FAST GEAR



Ponts à entrainement série HANDE

S201	CUMMINS ENGINE REPAIR (e.g., ISM, ISL)

Level, Mode and Duration	SENIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the Cummins engine following SHACMAN procedures. • **Disassemble** and **reassemble** the sub-assemblies (cylinder head, timing, injection). • **Check** the conformity of the components (dimensions, pressures). • **Test** and **adjust** the engine after repair.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Description and operation of the Cummins engine (Common Rail, Holset turbocharger , ECM). • Description of the lubrication and cooling system, • Description of the injection system, • Description of the electronic engine management system, • Removal and installation procedure following the manufacturer's instructions, • Disassembly and reassembly procedure following the manufacturer's instructions. • Component inspection procedure following the manufacturer's instructions. ► **Part Practical :** • CUMMINS engine according to the manufacturer's instructions. • CUMMINS engine according to the manufacturer's instructions. • Clean the components of the CUMMINS engine, • Check that components meet the manufacturer's expectations, • Replace non-compliant components to the manufacturer's expectations • Reassemble the engine components according to the manufacturer's instructions. • Reinstall the engine on the truck according to the manufacturer's instructions. • Test and adjust the engine on the truck. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technician performing the repair		
Prerequisites	Having acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, Cummins engine on bench, timing tool, INSITE™ diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined	To be defined	

S202	FULLER/FAST GEARBOX REPAIR

Level, Mode and Duration	SENIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Remove** and **refit** the gearbox according to the manufacturer's recommendations . • **Disassemble** and **reassemble** the gearbox and its synchronizers according to the manufacturer's instructions. • **Check for** wear on gears, bushings, synchronizers, • **Test and adjust** the gearbox after repair.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • e.g., RT-11509) or Fast Gear gearboxes used by SHACMAN . • Removal and installation procedure following the manufacturer's instructions, • Disassembly and reassembly procedure following the manufacturer's instructions. • Component inspection procedure following the manufacturer's instructions. ► **Part Practical :** • Submit the box of speeds following the manufacturer's instructions, • Disassemble the box of speeds following the manufacturer's instructions, • Clean the components of the box of speeds , • Check that components meet the manufacturer's expectations, • Replace non-compliant components to the manufacturer's expectations • Reassemble the gearbox components according to the manufacturer's instructions. • Rest the box of speeds on the truck following the manufacturer's instructions. • Test and adjust the gearbox of speeds on the truck. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technician performing the repair		
Prerequisites	Having acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Equipped training room and workshop, HW19710 gearbox on bench, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined		To be defined

S203	SHACMAN ELECTRICAL SYSTEM REPAIR

Level, Mode and Duration	SENIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Remove and reinstall** the components of the electrical system (starting, charging, air conditioning, and lighting circuits, etc.) in accordance with the manufacturer's recommendations. • **Disassemble and reassemble** electrical system components according to the manufacturer's recommendations. • **Check** that the components of the electrical systems (fuses, relays, starter, alternator, etc.) conform to the manufacturer's specifications. • **Test and adjust** electrical circuits according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • A review of the principles of electricity and electronics, • Description and operation of the electrical system and its various circuits (starting circuit, charging circuit, air conditioning circuit, lighting circuit, etc.) • Procedure for checking electrical components and parts according to the manufacturer's instructions. • Procedure for checking electrical components and parts according to the manufacturer's instructions. ► **Part Practical :** • Remove and install the electrical circuit components according to the manufacturer's recommendations. • Disassemble and reassemble the vehicle's electrical components according to the manufacturer's recommendations. • Clean the components and parts of the electrical circuit, • Check that the components of the electrical systems (fuses, relays, starter, alternator, etc.) conform to the manufacturer's specifications. • Replace non-compliant components according to the manufacturer's specifications. • Test and adjust the component in the vehicle's electrical circuit. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technician performing the repair		
Prerequisites	Having acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Equipped training room and workshop, starter, alternator, adapted vehicle(s) available.		
Date and place	To be defined		To be defined

S204	REPAIR OF CONVENTIONAL BRAKING SYSTEMS (WITH ABS/ASR)

Level, Mode and Duration	SENIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Remove** and **reinstall** the drum/disc brake mechanisms. • **Disassemble** and **reassemble** the components (wheel cylinders, valves). • **Check** the condition of the components. • **Test** and **adjust** the drum clearance and pneumatic system.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Description of the SHACMAN pneumatic braking system and ABS/ASR (WABCO or Knorr). • Description and operation of the S cam brake, • Removal and installation procedure of the cam brake S according to the manufacturer's instructions • Disassembly and reassembly procedure for the S cam brake according to the manufacturer's instructions. • Procedure for checking the components of the S cam brake according to the manufacturer's instructions. ► **Part Practical :** • Remove and reinstall a relay valve, following the manufacturer's recommendations. • Disassemble and reassemble the parking brake relay valve according to the manufacturer's instructions. • Remove the cam brake mechanism S according to the manufacturer's instructions. • Disassemble the S cam brake mechanism according to the manufacturer's instructions. • Clean the components of the S cam brake mechanism, • Check that components meet the manufacturer's expectations, • Replace components that do not meet the manufacturer's specifications. • Reassemble the components of the S cam brake mechanism according to the manufacturer's instructions. • Reinstall the cam brake mechanism S according to the manufacturer's instructions. • Test and adjust the S cam brake. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technician performing the repair		
Prerequisites	Having acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined		To be defined

S205	HANDE BRIDGE AND GEARBOX REPAIRS

Level, Mode and Duration	SENIOR	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Place** and **rest** the axle and wheel reducers, according to the manufacturer's recommendations, • **Disassemble** and **reassemble** the vehicle's axle and wheel reducers according to the manufacturer's recommendations. • **Control** compliance components of the organs (bridge and gearboxes) according to the manufacturer's specifications, • **Test** and **adjust** the components of the bridge and wheel reducers according to the manufacturer's recommendations.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Description and operation of the bridge and gearboxes, • Removal and installation procedure of the bridge and gearboxes, following the manufacturer's instructions, • Disassembly and reassembly procedure for the bridge and gearboxes, following the manufacturer's instructions. • Procedure for checking bridge components and gearboxes, following the manufacturer's instructions. ► **Part Practical :** • Remove the bridge and gearbox mechanism, following the manufacturer's instructions . • Disassemble the bridge and gearbox mechanism, following the manufacturer's instructions. • Clean the components of the bridge mechanism and the gearboxes, • Check that the bridge mechanism components and gearboxes conform to the manufacturer's specifications. • Replace components that do not meet the manufacturer's specifications. • Reassemble the bridge mechanism and gearbox components, following the manufacturer's instructions. • Reinstall the bridge and gearbox mechanism, following the manufacturer's instructions. • Test and adjust the bridge and gearbox mechanism . ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technician performing the repair		
Prerequisites	Having acquired the skills of a JUNIOR Technician (Preventive Maintenance)		
Language	French		
Equipment	Training room and workshop equipped, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined	To be defined	

EXPERT TECHNICIAN

DIAGNOSIS



S301	CUMMINS ENGINE DIAGNOSTICS AND ARCHITECTURE

Level, Mode and Duration	EXPERT	Face-to-face	5 Days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Diagnosing** complex faults on the engine and electronic architecture. • **Use** the INSITE™ diagnostic tool and oscilloscope in an advanced way. • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Advanced reminder of the Common Rail system and electronic management (ECM). • Study of the SHACMAN CAN network and communications between modules. • 7-step diagnostic methodology applied to complex faults. ► **Part Practical :** 1. Workshops on simulated breakdowns: loss of power, excessive smoke, communication failure. 2. Analysis of parametric data and graphs with INSITE™. 3. Using the oscilloscope to analyze sensor signals (crankshaft , camshaft , rail pressure). ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Expert diagnostic technicians.		
Prerequisites	Having acquired the skills of Senior level (S201).		
Language	English - French		
Equipment	Equipped training room and workshop, INSITE™ diagnostic tool, oscilloscope, SHACMAN vehicles with simulated faults, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined	To be defined	

S302	DIAGNOSTIC OF FULLER/FAST GEAR TRANSMISSION AND CONTROLS

Level, Mode and Duration	EXPERT	Face-to-face	05 days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Diagnosing** faults related to conventional braking systems and ABS/ASR. • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Reminder on the description and operation of the components of the conventional braking system and ABS/ASR (WABCO/Knorr). • Malfunctions of components (valves, tanks, wheel sensors, etc.). • The diagnostic steps and features of the SHACMAN Diag tool. ► **Part Practical :** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diag tool). 2. Formulate hypotheses about the cause of the malfunction/breakdown. 3. Test or check the conformity of the braking system's systems and components with the manufacturer's recommendations. 4. Identify the most likely hypothesis based on the tests and symptoms. 5. Develop a recommendation for the components to be changed or repaired. 6. Perform the repair of the defective component. 7. Test and adjust the system involved in the truck repair. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a SENIOR Technician (Repair) - Module S204		
Language	French		
Equipment	Equipped training room and workshop, SHACMAN diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined	To be defined	

S303	ADVANCED DIAGNOSTIC OF ELECTRICAL AND ELECTRONIC SYSTEMS

Level, Mode and Duration	EXPERT	Face-to-face	05 days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Troubleshooting** complex electrical faults (intermittent, multiple). • **Diagnosing** communication faults on CAN J1939 and LIN networks. • **Isolate** a faulty electronic module within the vehicle's network architecture. • **Mastering** the diagnosis of comfort and safety systems (air conditioning, basic ADAS).		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • Complete electronic architecture of a SHACMAN truck (Engine ECU, ABS, Gearbox, Instrumentation, etc.) and network topology. • Advanced principles of CAN (J1939) and LIN networks: signals, termination, troubleshooting. • Study of complex electrical diagrams and manufacturer diagnostic trees ► **Part Practical :** 1. Advanced use of the SHACMAN diagnostic tool 2. Network diagnostics with oscilloscope/analyzer **3.** Complex simulated fault resolution ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Expert technicians in electrical diagnostics. Must have acquired the skills of Senior level (S203).		
Prerequisites	Having acquired the skills of a SENIOR Technician (Repair)		
Language	French		
Equipment	Equipped training room and workshop , SHACMAN diagnostic tool , oscilloscope/network analyzer, high-precision multimeter, vehicles with simulated faults, detailed wiring documentation.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined	To be defined	

S304	S304 - BRAKE SYSTEM AND ABS/ASR DIAGNOSTIC

Level, Mode and Duration	EXPERT	Face-to-face	05 days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Diagnose** malfunctions related to the conventional braking system • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • A reminder of the description and operation of the components of a conventional braking system, • conventional braking system , • The diagnostic steps and features of the Diag tool. ► **Part Practical :** 1. Collect the various symptoms of the malfunction/breakdown and the history of breakdowns and repairs of the truck (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. To test or verify the conformity of the systems and components of the conventional braking system with the manufacturer's recommendations, 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components to be replaced or repaired, 6. Perform the repair of the defective component, 7. Test and adjust the system involved in the truck repair. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a SENIOR Technician (Repair)		
Language	French		
Equipment	Equipped training room and workshop, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined	To be defined	

S305	DIAGNOSTIC OF BRIDGES AND GEARBOXES FOR HANDE

Level, Mode and Duration	EXPERT	Face-to-face	05 days
Presenter	**SINOTRUCK Expert Technical Trainer**		
Goals	• **Diagnose** malfunctions related to the operation of the bridge and gearboxes, • **Recommend** remedial actions.		
Description Theory (40%) Practical (60%)	► **Part Theoretical :** • HanDe series drivetrain and bridges • HanDe series bridges and gearboxes • Malfunctions in bridge and gearbox systems (noise, vibration, play) • The diagnostic steps and features of the Diag tool. ► **Part Practical :** 1. Collect the various symptoms of the malfunction/failure and the history of breakdowns and repairs of the forklift (Customer and Diagnostic tool), 2. Formulate hypotheses regarding the cause of the malfunction/failure, 3. To test or verify the conformity of the system and components with the manufacturer's recommendations. 4. Identify the most likely hypothesis based on the tests and symptoms, 5. Develop a recommendation for the components/parts to be changed or repaired, 6. Carry out the repair of the defective component or organ, 7. Test and adjust the system concerned by the repair. ► **section :** • Initial Test (Prerequisite Assessment) • Final Test (Evaluation of Acquired Skills)		
Target audience	Technicians performing diagnostics		
Prerequisites	Having acquired the skills of a SENIOR Technician (Repair) Module S205		
Language	French		
Equipment	Equipped training room and workshop, diagnostic tool, adapted vehicle(s) available.		
Documentation	Attendance sheet, satisfaction sheet, certificate of participation.		
Date and place	To be defined with the client	To be defined	